DATA SCIENCE CAPSTONE PROJECT

SpaceX Falcon 9 First Stage Landing Prediction

Final Presentation

GitHub: [INSERT ACTUAL GITHUB URL]

Executive Summary

- Collected launch data using the SpaceX REST API and Wikipedia web scraping.
- Cleaned and engineered features using Pandas/NumPy.
- Performed EDA with Python visualization and SQL.
- Built interactive Folium maps and a Plotly Dash dashboard.
- Compared Logistic Regression, SVM, Decision Tree and KNN.
- Logistic Regression, SVM and KNN each achieved 83.33% test accuracy and Jaccard 0.80.

Introduction & Problem Statement

- Reusable Falcon 9 first stages are a major source of launch-cost savings.
- Landing success is a critical operational outcome.
- Goal: predict whether the first stage will successfully land.
- Study factors such as flight number, payload, orbit, launch site and booster history.

Data Collection — SpaceX API

Jupyter Notebook → REST API → GET request → JSON response → field selection → Pandas DataFrame

Collected launch, payload, orbit, landing, booster and geographic information.

Notebook: 01-Data Collection API Lab.ipynb

GitHub: [INSERT ACTUAL URL]

Data Collection — Web Scraping

Jupyter Notebook → Wikipedia → HTTP request → HTML/table extraction → Pandas DataFrame

Used as a complementary historical source.

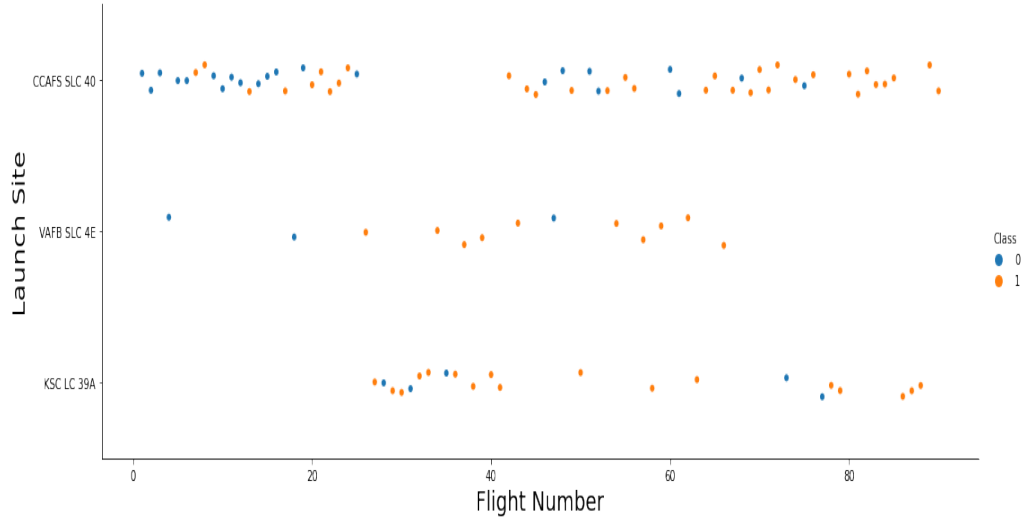
Notebook: 02-Data Collection with Web Scraping lab.ipynb

GitHub: [INSERT ACTUAL URL]

Data Wrangling Methodology

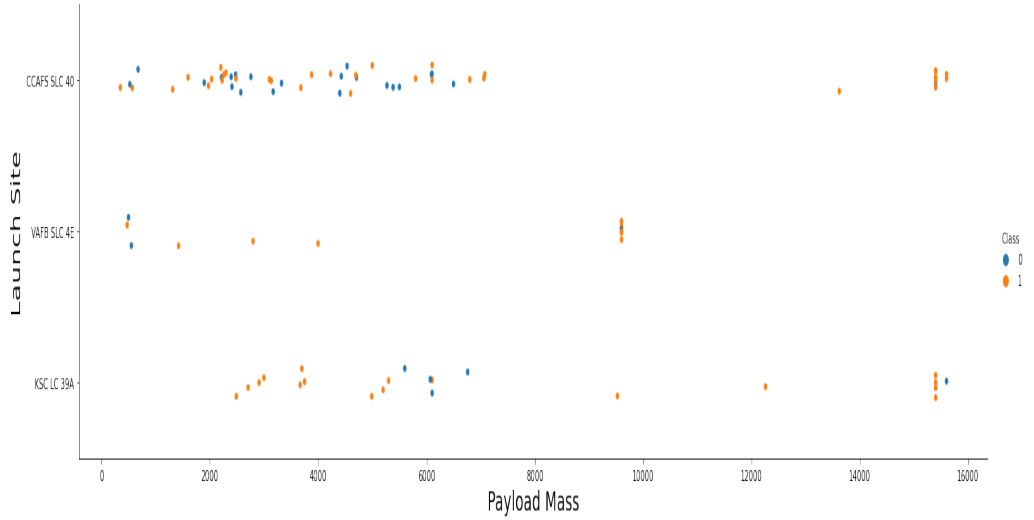
- Inspect and clean raw records.
- Handle missing/unnecessary fields.
- Create binary landing class.
- One-hot encode categorical features.
- Standardize numerical predictors.
- Split data into training and testing sets.

EDA — Flight Number vs Launch Site



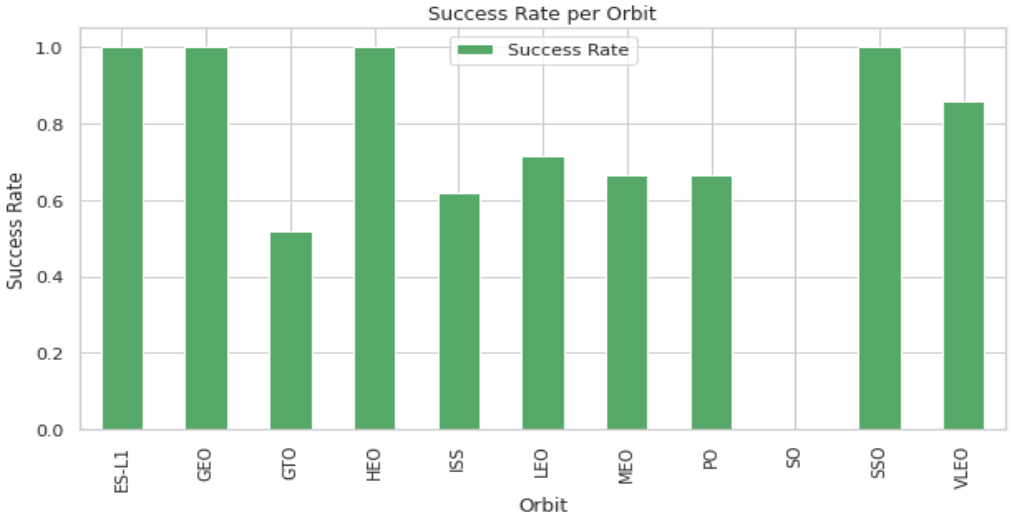
CCAFS SLC 40: 55 trials, about 60% success. KSC LC-39A and VAFB SLC 4E: about 77% success.

EDA — Payload vs Launch Site



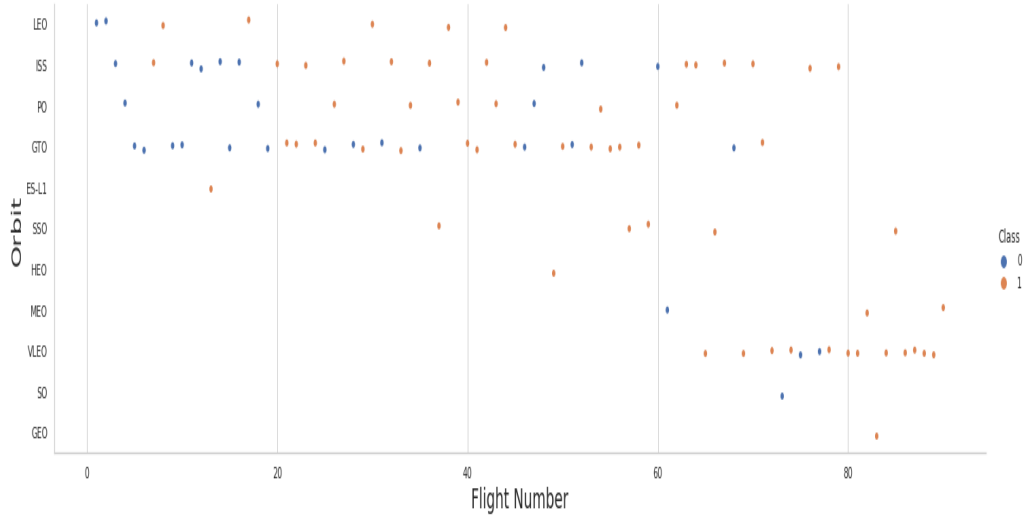
No strong direct relationship between payload mass and landing success is visible when grouped only by launch site.

EDA — Success Rate by Orbit



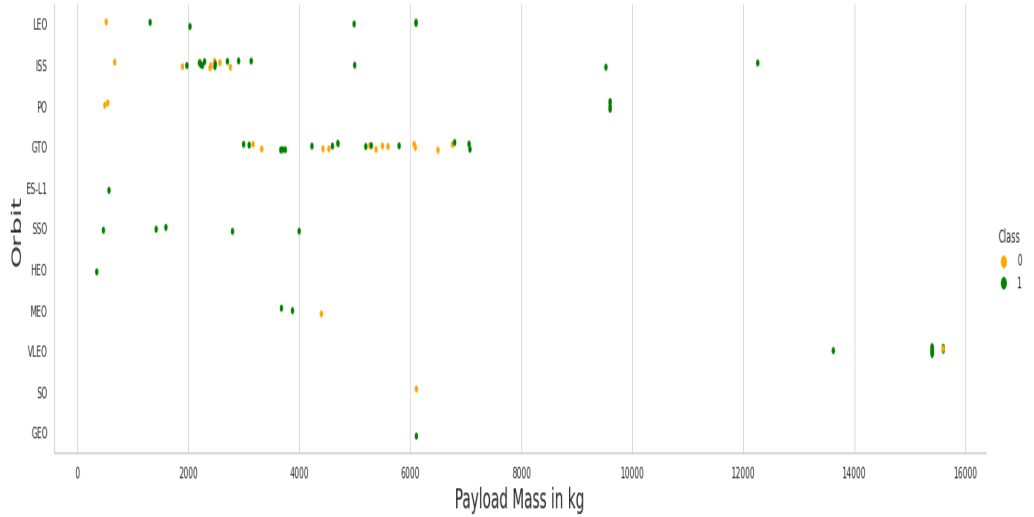
ES-L1, GEO, HEO and SSO perform strongly; GTO is comparatively weak in the notebook analysis.

EDA — Flight Number vs Orbit



LEO shows an improving success pattern as flight number increases.

EDA — Payload vs Orbit



Payload effects differ across orbit types, supporting a multivariate modeling approach.

EDA — Yearly Launch Success Trend



Success rate rises substantially over time, reaching 90% in 2019 and 84.21% in 2020.

EDA with SQL — Main Questions

- Unique launch sites
- Sites beginning with CCA
- NASA (CRS) payload totals
- Average payload for F9 v1.1
- First successful ground landing
- Successful drone-ship boosters by payload range
- Mission outcome counts
- Maximum payload booster versions
- 2015 failed drone-ship landings
- Ranked landing outcomes

SQL Results — Key Numbers

4 distinct launch sites

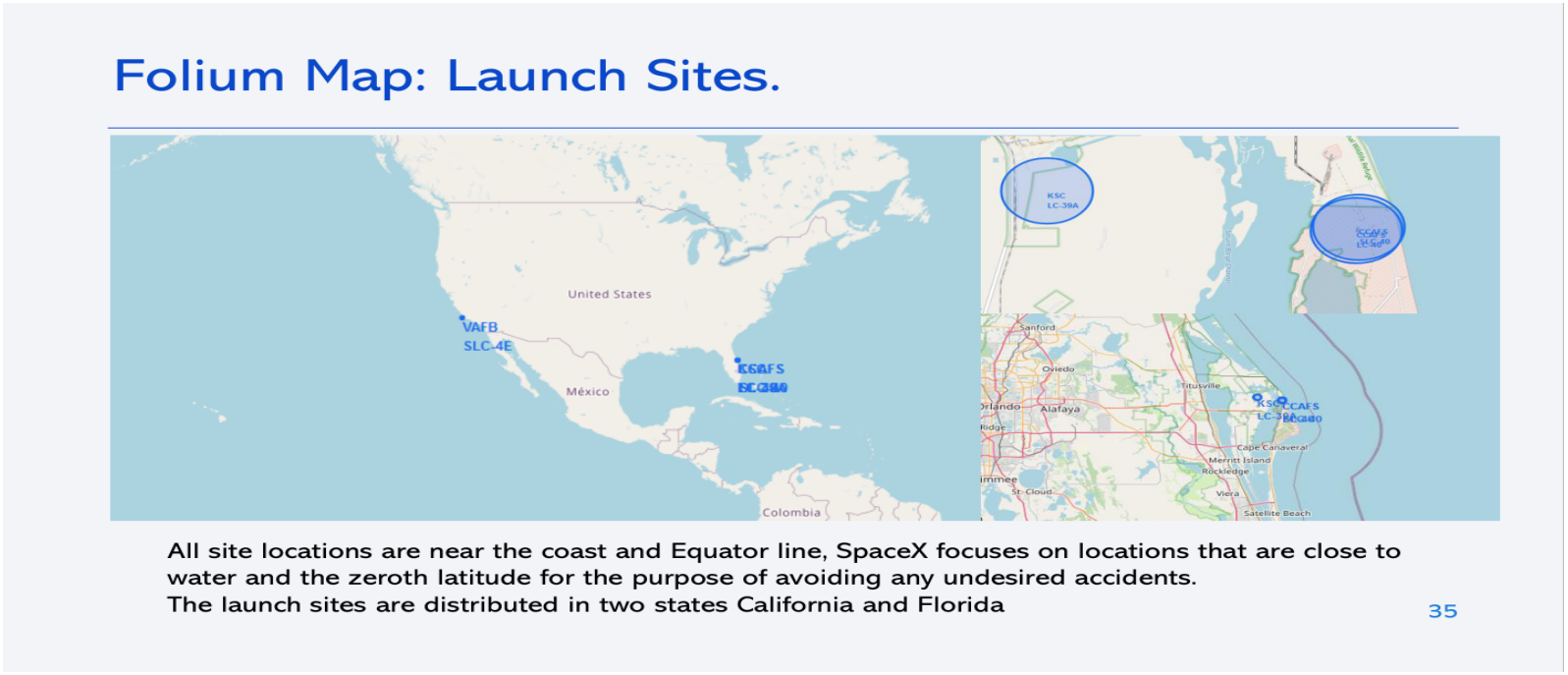
NASA (CRS) payload total ≈ 45,596 kg

F9 v1.1 average payload ≈ 2,928 kg

First successful ground landing: 22 December 2015

Mission outcomes: 99 Success; 1 Failure (in flight); 1 Success (payload status unclear)

Folium — Launch Site Map



Four launch sites are mapped interactively. Sites are close to coastlines and low-latitude regions.

Folium — Success/Failure Markers



Green markers represent successful landings; red markers represent failures. Marker clustering reduces overlap.

Folium — Proximity Analysis

Folium Map: Closest Proximities to CCAFS LC-40





Proximities Cordinates

	Location	Lat	Long
0	Orlando_Location	28.52300	-81.38260
1	Coastline_Location	28.56146	-80.56746
2	Highway_Location	28.56270	-80.58703

we calculated the distances between the launch site (CCAFS LC-40) to its proximities

Orlando City Distance ≈ 78.8 Km,
Coastline Distance ≈ 0.97 Km,
Highway Distance ≈ 0.95Km

37

CCAFS LC-40 example: Orlando ≈ 78.8 km; coastline ≈ 0.97 km; highway ≈ 0.95 km.

Plotly Dash — Interactive Dashboard



Launch-site dropdown + success/failure summary + interactive payload/outcome analysis.

Plotly Dash — Payload vs Outcome

<Dashboard Screenshot 3>



This interactive scatter plot shows the launch outcome based on the relationship between the payload mass and the final outcomes as we can infer that the success rate per booster version for example, the payload mass < 4000 kg more likely to be successful.

Interactive scatter plot helps explore payload mass against landing outcome.

Predictive Analysis — Machine Learning

- Target: landing success (1) vs unsuccessful landing (0).
- 80/20 train-test split.
- Standardization and one-hot encoding.
- GridSearchCV used for hyperparameter tuning.
- Models: Logistic Regression, SVM, Decision Tree, KNN.

Model Evaluation

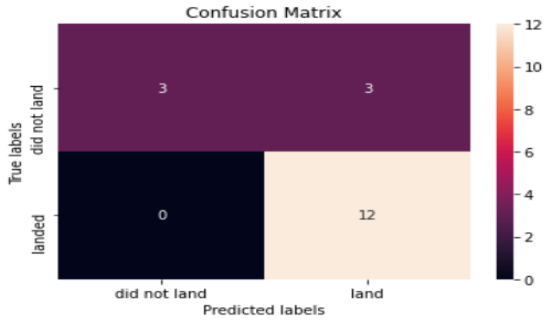
Logistic Regression — validation 84.64%; test 83.33%; Jaccard 0.80; macro F1 0.7778.

SVM — validation 84.82%; test 83.33%; Jaccard 0.80; macro F1 0.7778.

Decision Tree — validation 88.93%; test 77.78%; Jaccard 0.7333; macro F1 0.7231.

KNN — validation 84.82%; test 83.33%; Jaccard 0.80; macro F1 0.7778.

Confusion Matrix




```
Logistic Regression:
Jaccard Score of = 0.8
F1 Score = 0.7777777777777778
====================================
SVM:
Jaccard Score of = 0.8
F1 Score = 0.7777777777777778
====================================
Decision Tree:
Jaccard Score of = 0.6666666666666666
F1 Score = 0.6727272727272727
====================================
KNN:
Jaccard Score of = 0.8
F1 Score = 0.7777777777777778
====================================
```

Confusion Matrix

Logistic Regression , SVM and KNN have the same confusion matrix and results:

- True Positive = 12
- False Positive = 0
- True Negative = 3
- False Negative = 3

Top-performing models: TP = 12, FP = 0, TN = 3, FN = 3.

Best Model / Final Selection

- Logistic Regression, SVM and KNN tie on the held-out test metrics.
- All three achieve 83.33% test accuracy.
- All three achieve Jaccard = 0.80 and macro F1 = 0.7778.
- Decision Tree has stronger cross-validation but weaker held-out test performance.
- Logistic Regression is a practical final choice because it is simple and interpretable.

Key Insights

- Launch site strongly influences observed landing success.
- Flight number is associated with improving success.
- Payload effects depend on orbit and other launch conditions.
- GTO is comparatively difficult for first-stage recovery.
- Landing success improves strongly over time.
- Geospatial proximity to coast and transportation infrastructure is evident.

Conclusion

- An end-to-end data science workflow successfully analyzed Falcon 9 landing outcomes.
- API + web scraping created the project dataset.
- EDA and SQL revealed operational patterns.
- Folium and Plotly Dash made the findings interactive.
- Logistic Regression, SVM and KNN were the strongest test-set performers.
- Historical launch attributes provide useful predictive information for landing success.

GitHub & Submission Checklist

- Final Report PDF — prepared separately.
- Final Presentation PDF — this file.
- GitHub URL — replace the placeholder with your public repository URL.
- Ensure notebooks, Dash app and CSV are present in the repository.
- Upload the two PDFs separately according to the 15-point rubric.